Relevant Projects & Activities

Tom Gilgenkrantz

Tech

March 24, 2025

Contents

1 Personal Website Development **2**

2 Board Game **2**

3 Contact Management Software **3**

4 Group classification using decision tree **4**

5 Crowd simulation **4**

1 Personal Website Development

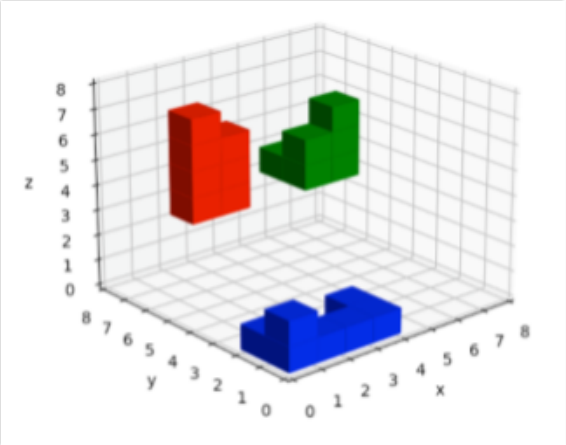
HTML, CSS, Javascript

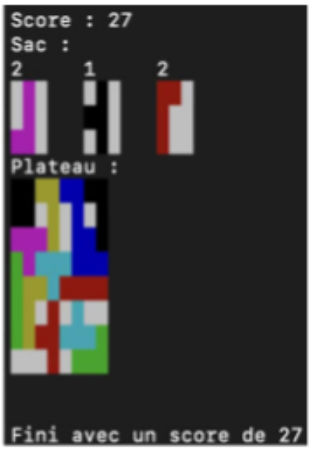
- Creation of a personal HTML website (Portfolio + Online Courses)
- Animation with CSS/JavaScript
- Development of a responsive web design
- Deployment of the website on a cloud hosting platform (github)
- User Database Management with Firebase
- Review Scraping with Python (requests)

2 Board Game

Python, C

- Implementation in C of a 2D Tetris-like game
- Implementation in Python of a 3D game
- Mathematical research documents on these games
- Research activities on automatic game algorithm

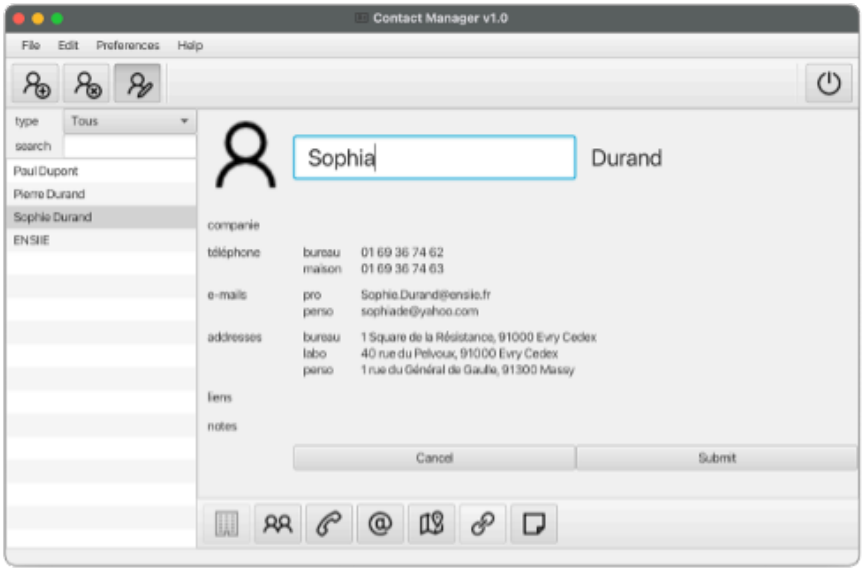




3 Contact Management Software

Java

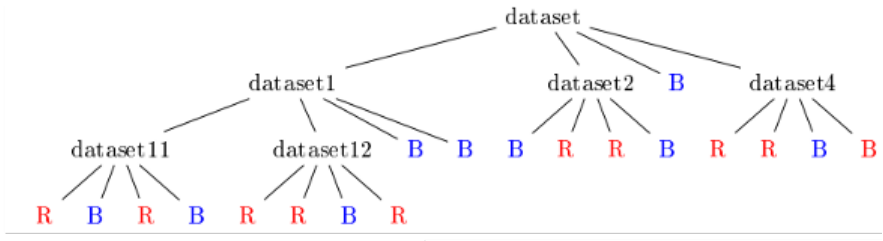
- Application to manage contacts in an object-oriented language

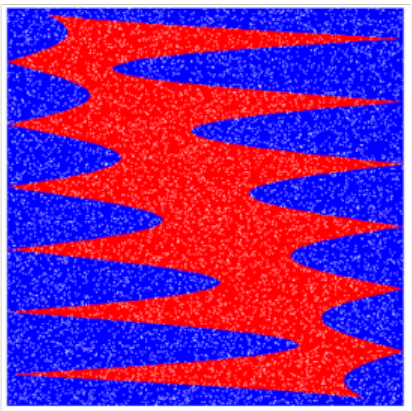


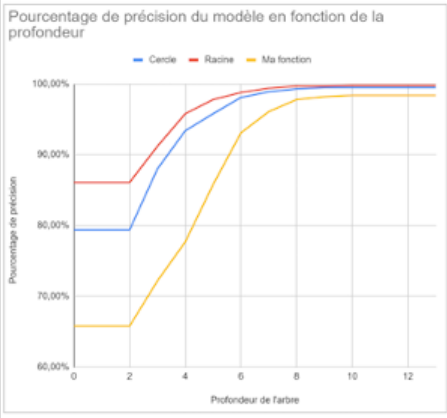
4 Group classification using decision tree

Ocaml

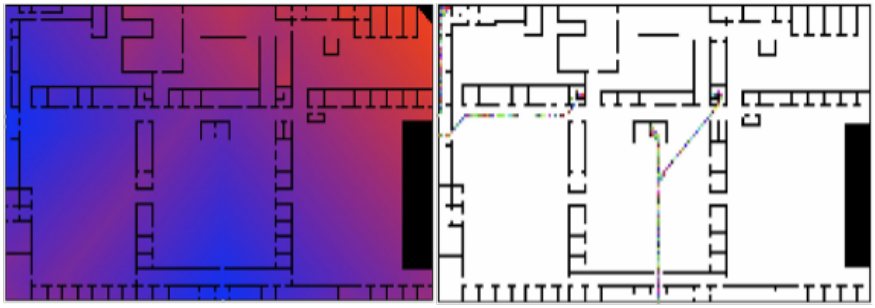
- Classification algorithm for datasets using quaternary trees
- Research documents on model performance







5 Crowd simulation

Python

- Use of Dijkstra and adaptation of A-star to manage people's behavior
- Development of a cellular automaton model
- Application to my high school and proposal to improve its emergency exits